



Assessor's Office Open Data: What's public, and what for

Cook County Bureau of Economic Development
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Internal shoutout: William Ridgeway, Senior Data Scientist
External shoutouts: Josh Kalov (GIS) and Jadie Oldfield (BOT)

Today



Mission and role of the CCAO



Why Open Data



A Tour of CCAO's Open Data assets



Questions? Requests?



Demo?

What the Assessor's Office does

1 Value property and 2 administer tax-saving exemptions and incentives.

Cook County Assessor's Office

- Revaluation and reassessment
- Assessment appeals
- Property tax incentives and exemptions
- New Property

*separately elected

Not the Cook County Assessor's Office

- Tax bills
- Tax rates
- Tax revenues
- Who is eligible for exemptions
- Create or expire TIF districts
- Approve permits
- Zoning and development
- Insurance rates
- Rents
- The real estate market

Open data boosts Cook County's profile

Hidden Errors in Big Data: The Case of Property Records

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Researchers found that our data is **higher quality** than that of two large commercial brokers who sell data. Quotes (emphasis mine):

“**Cook County is an outlier in terms of the amount and quality** of property data that it makes available for researchers to download and use.”

“Our findings underscore the importance of **accessible open data** from federal, state, and local governments.”



Why Open Data

The “Tax Assessor” can be unpopular. Why is Open Data worth the effort and risk?

Public have a right to “raw” data.

Property assessment is a public function. Parcel-level data should be visible, free, and clearly documented.

Open data helps the institution.

It reduces FOIA requests, makes data requests easier for us to respond to, and allows the public, journalists, and researchers to conduct analyses.

Open data powers important work.

Our Open Data users have published research papers in academic journals, created web apps, and identified errors for us to fix.



README.md

Hi there! 🙋

We're the [Cook County Assessor's Data Department](#). We make predictions for every [residential](#) and [condominium](#) property in Cook County. We aim to make the property system more transparent, predictable, and fair.

Our work

Our tiny team has done some big work since 2018. Here are some of our projects:

- [model-res-avm](#) 🏠 - Model to create assessed values for all single-family homes in Cook County
- [model-condo-avm](#) 🏢 - Predictive model for all residential condominiums
- [ptaxsim](#) 🇺🇸 - Database and R package to estimate counterfactual property tax
- [data-architecture](#) 🗑️ and [documentation](#) 📖 - ETL/ELT stuff, SQL, and more
- [Cook County Open Data](#) 📊 - Public datasets, including historical data

Get involved

- Check out the [public](#) repo for [data requests](#) 🔍, our [talks](#) 🗣️, and [other resources](#) 📁
- View all our open [issues](#) ✅ and [pull requests](#) 🚀. We welcome contributions!
- Visit our [wiki](#) 📖 to learn more about the Data Department's mission and vision

github.com/ ccao-data

- Open Data assets
- Documentation
- SOPs
- Code
- Automated refresh



A Tour of CCAO Open Data

Cook County Open Data Portal

Assessed Values ★

Assessed values of every parcel for each tax year in Cook County dating back to 1999.

Each row is one:

PIN, per tax year.

Example use case:

“What were the top 10 highest assessed values in Bloom in 2023?”

Assessor - Assessed Values					
pin	tax_year	class	townsh...	townsh...	neighb...
3221321009	2020	202	12	Bloom	12111
3221403005	2020	EX	12	Bloom	12111
3221404037	2020	EX	12	Bloom	12111
3221411008	2020	EX	12	Bloom	12111
3221418003	2020	EX	12	Bloom	12111
3222104015	2020	890	12	Bloom	12150

Cook County Open Data Portal

Parcel Universe (Historical) ★ & Parcel Universe (Current Year Only) ★

Spatial data, such as tax districts, Chicago Community Area, and census tract, for every parcel in Cook County.

Each row is one:

PIN, either per historical tax year or for this tax year.

Example use cases:

“Which PINs are in Chicago’s 3rd ward?”

“Where is all the vacant land in Cook County?”

Columns (124)

Column Name	Description
Tt ward_num	Ward number
# ward_chicago_data_year	Chicago ward data year
# ward_evanston_data_year	Evanston ward data year
Tt chicago_community_area_num	Chicago community area number
Tt chicago_community_area_name	Chicago community area name

Cook County Open Data Portal

Parcel Addresses

Mailing and situs addresses for parcels.

Each row is one:

PIN, per tax year.

Example use case:

“What are the physical and mailing addresses of every property in Maywood?”

 property_address	Property street address
 property_city	Property city
 property_state	Property state
 property_zip	Property zip code
 mailing_name	Taxpayer mailing name
 mailing_address	Taxpayer mailing street address
 mailing_city	Taxpayer mailing city
 mailing_state	Taxpayer mailing state
 mailing_zip	Taxpayer mailing zip code
 owner_name	Owner mailing name
 owner_address	Owner mailing street address

Cook County Open Data Portal

Permits

Historical permit data submitted by municipalities to the Assessor's Office, with extra metadata from processing.

Each row is one:

PIN associated with a permit issued on a specific date.

Example use cases:

“How many permits have been received by the CCAO in Berwyn?”

Columns (22)

Column Name	Description
 pin	Parcel Identification Number (PIN)
 permit_year	Year municipality issued permit to applicant
 permit_number	Permit number created during permit submission to CCAO
 local_permit_number	Permit number recorded by reporting municipality
 date_issued	Date municipality issued permit to applicant
 estimated_date_of_completion	Estimated date permitted work will be complete
 status	Status of permit in CCAO's permit workflow
 assessable	Whether permitted work will affect property's assessed value
 amount	Amount permitted work is estimated to cost
 municipality	Taxing district name, as seen on Cook County tax bills

Cook County Open Data Portal

Single- and Multi-Family Characteristics

Property characteristics of single and small multi-family buildings.

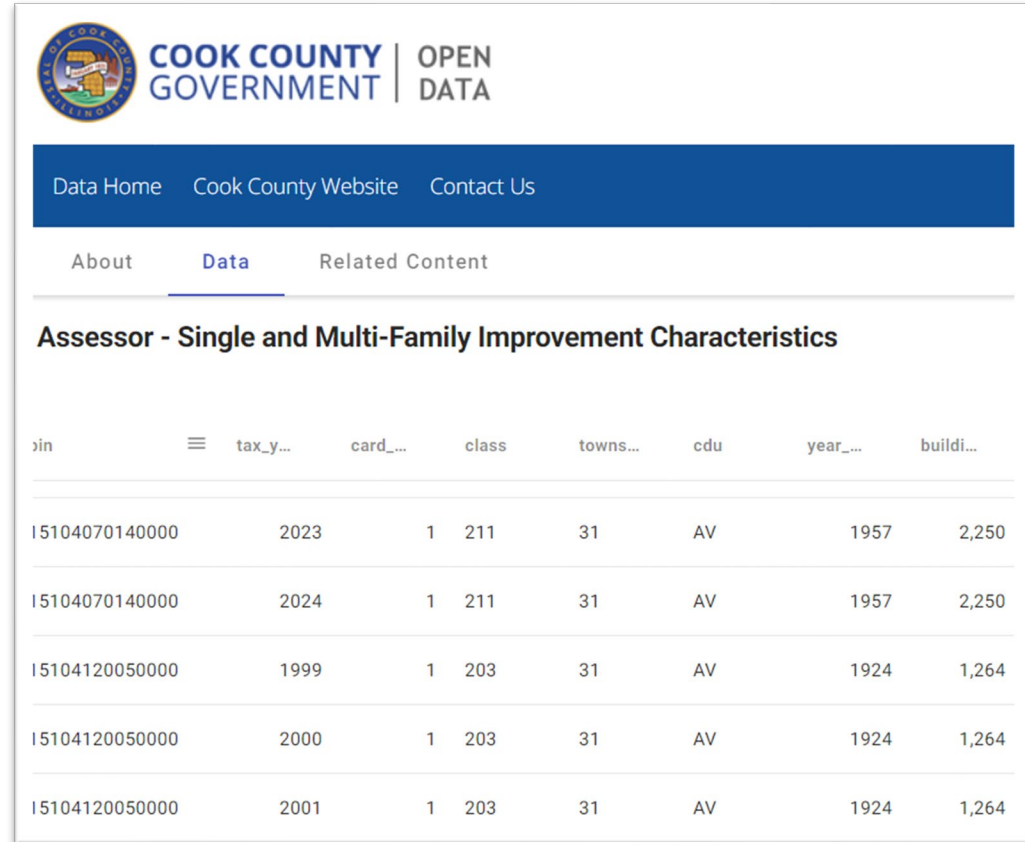
Each row is one:

Improvement 🏠, per PIN, per tax year.

Example use cases:

“What is the PIN for every home in Norwood Park under 1200 sf?”

“What’s the oldest home in Cook County?”



The screenshot shows the Cook County Open Data Portal interface. At the top is the Cook County Government logo and 'OPEN DATA' text. Below is a navigation bar with links: Data Home, Cook County Website, and Contact Us. A secondary navigation bar has links: About, Data (selected), and Related Content. The main heading is 'Assessor - Single and Multi-Family Improvement Characteristics'. Below this is a table with columns: pin, tax_y..., card_..., class, towns..., cdu, year_..., and buildi... The table contains five rows of data.

pin	tax_y...	card_...	class	towns...	cdu	year_...	buildi...
15104070140000	2023	1	211	31	AV	1957	2,250
15104070140000	2024	1	211	31	AV	1957	2,250
15104120050000	1999	1	203	31	AV	1924	1,264
15104120050000	2000	1	203	31	AV	1924	1,264
15104120050000	2001	1	203	31	AV	1924	1,264

Cook County Open Data Portal

Residential Condominium Unit Characteristics

Characteristics of condominium units used in the condo model.

Each row is one:

Condo* PIN, per tax year.

Example use case:

“Which Chicago condo has the highest square footage?”**

** includes non-livable spaces*

*** when data is available*

Assessor - Residential Condominium Unit Characteristics

ain	pin10	card_...	tax_y...	class	towns...	prorat...	pin_pr...
010110103	010110103	1	2021	299	10	010110103	0.00217
010110103	010110103	1	2022	299	10	010110103	0.00217
010110103	010110103	1	2023	299	10	010110103	0.00217
020110001	020110001	1	2026	299	29	020110001	0.0026062
010110103	010110103	1	2007	299	10	010110103	0.00217

Cook County Open Data Portal

Commercial Valuation Data

Data used to value commercial properties.

Each row is one:

Commercial property, per tax year. Commercial properties are often multi-parcel.

Example use case:

“What commercial buildings in Oak Park are over 100,000 sf?”

Assessor - Commercial Valuation Data

keypin	pins	year	towns...	model...	class(...
01-02-202-C	01-02-202-C	2022	Barrington	5-17s	5-17
01-01-201-C	01-01-201-C	2022	Barrington	5-17s	5-17
01-33-305-C	01-33-305-C	2022	Barrington	5-17s	5-17

Note: a KeyPIN only has a row in its reval tax year. A Barrington KeyPIN only exists in 2022, 2025...

Cook County Open Data Portal

Parcel Sales

Parcel sales for real property transactions in Cook County.

Each row is one:

PIN associated with a sale.

Example use case:

“What are all the commercial property transactions in Chicago last year?”

Assessor - Parcel Sales					
pin	year	towns...	neigh...	class	sale_d...
2022326016	2023	70	70030	318	January 12,
2515117016	2023	70	70220	203	January 18,
2002302036	2023	70	70010	205	January 19,
2011400016	2023	70	70020	299	January 10,
2035209016	2023	70	70080	203	January 25,
2503221016	2023	70	70120	211	January 26,

Cook County Open Data Portal

Appeals

Each row is one:

Appealed PIN, per appeal,
per tax year.

Example use case:

“In tax year 2025, how many
condo appeals got a no
change result, and why?”

Note: CCBOR also publishes appeal data.

Assessor - Appeals					
pin	tax_y...	class	towns...	case_...	heari...
142131405	2025	299	73	25-73-1754	Current yea
142131405	2000	299	73	6661	Current yea
142131405	2003	299	73	11184	Current yea
142131405	2021	299	73	21-73-4717	Current yea
142131405	2022	299	73	22-73-6535	Current yea

Cook County Open Data Portal

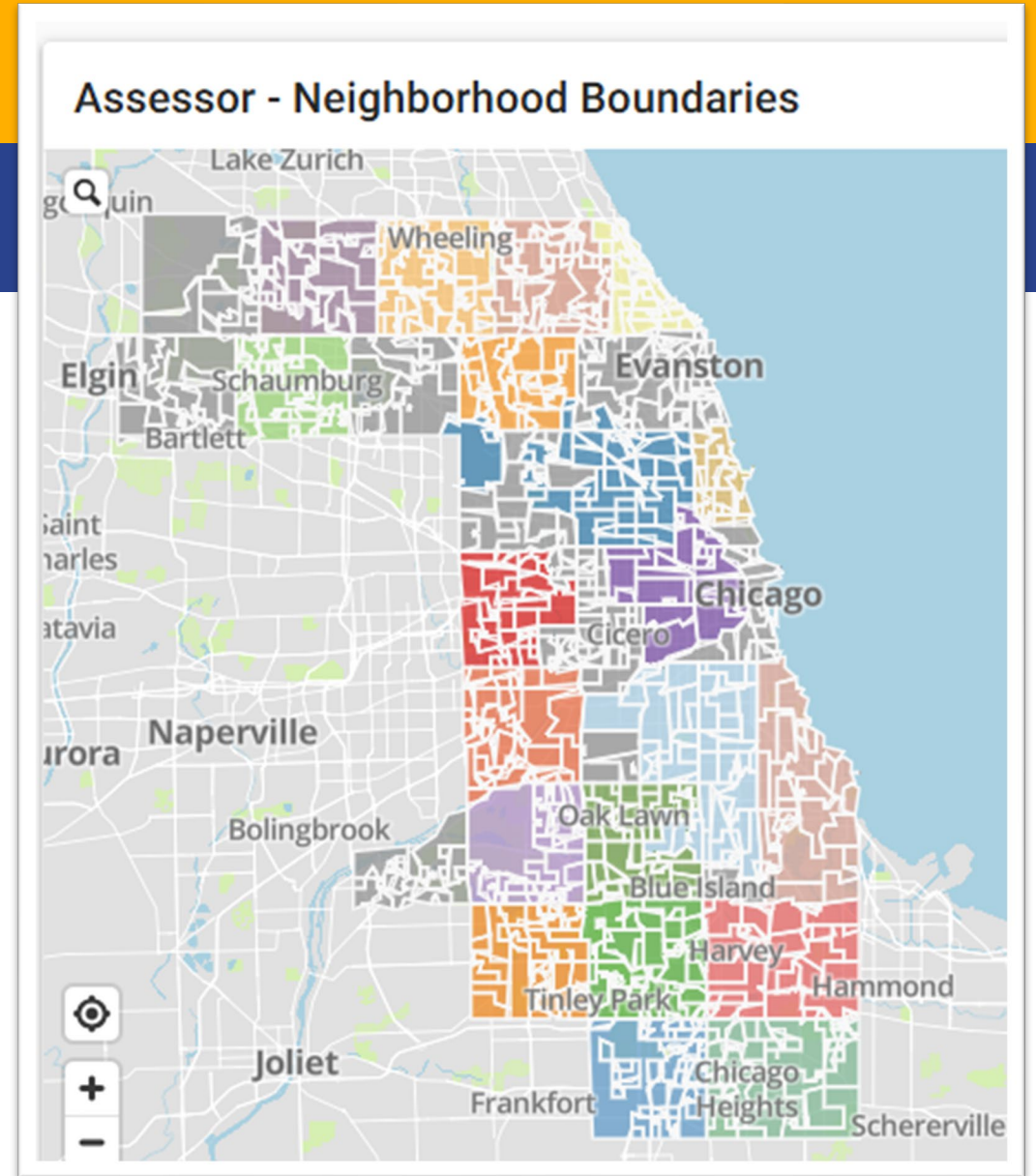
Neighborhood Boundaries

Each row is one:

Residential neighborhood code.

Example use case:

Thematic mapping!



Cook County Open Data Portal

Property Tax-Exempt Parcels

Parcels with property tax-exempt status.

Each row is one:

Tax-exempt PIN, per tax year.

Example use case:

“What are all the tax-exempt, County-owned properties in Niles?”

Tt	pin	Parcel Identification Number (PIN)
#	tax_year	Tax Year
Tt	township_name	Township name
Tt	township_code	Township code
Tt	owner_num	Owner number
Tt	owner_name	Owner name
Tt	class	Class
Tt	property_address	Property street address
Tt	property_city	Property city
#	longitude	Parcel centroid longitude
#	latitude	Parcel centroid latitude

Cook County Open Data Portal

Parcel Proximity ★

10-digit parcels and their proximity to various spatial features, such as train stops and Lake Michigan, per year.

Each row is one:

10-digit PIN, per tax year.

Example use case:

“In 2024, which 10-digit parcels are within 500 feet of a highway?”

Tt	nearest_cta_stop_name	Nearest CTA stop name
Tt	nearest_cta_stop_dist_ft	Nearest CTA stop distance (feet)
#	nearest_cta_stop_data_year	Nearest CTA stop data year
Tt	nearest_golf_course_id	Nearest golf course ID
Tt	nearest_golf_course_dist_ft	Nearest golf course distance (feet)
#	nearest_golf_course_data_year	Nearest golf course data year
Tt	nearest_hospital_gnis_code	Nearest hospital GNIS code
Tt	nearest_hospital_name	Nearest hospital name
Tt	nearest_hospital_dist_ft	Nearest hospital distance (feet)
#	nearest_hospital_data_year	Nearest hospital data year
Tt	lake_michigan_dist_ft	Lake Michigan distance (feet)



Demo!

How to filter, query, download, and share data.

Cook County Open Data Portal

Commercial Valuation Data

Each row is one:

KeyPIN, per tax year*.

A KeyPIN is the primary PIN for group of PINs assessed as one economic unit.

Example use case:

“What commercial buildings in Oak Park and River Forest have a bldg. sf over 100,000 sf?”

Assessor - Commercial Valuation Data

keypin	pins	year	towns...	model...	class(...
01-02-202-C	01-02-202-C	2022	Barrington	5-17s	5-17
01-01-201-C	01-01-201-C	2022	Barrington	5-17s	5-17
01-33-305-C	01-33-305-C	2022	Barrington	5-17s	5-17

Note: a KeyPIN only has a row in its reval tax year. A Barrington KeyPIN only exists in 2022, 2025...

Step 0: Explore! Can I answer this using Open Data?

“What commercial buildings in Oak Park and River Forest have a bldg. sf over 100,000 sf?”

Columns (76)

Column Name	Description	API Field Name	Data Type
 keypin	Parcel identification number (PIN). Primary PIN used for valuing properties associated with multiple PINs.	keypin	Text
 pins	All PINs associated with primary PIN that should be valued collectively	pins	Text
 year	Year	year	Number
 township	Township name	township	Text
⋮			
 bldgsf	Building square footage for key PIN and all associated PINs	bldgsf	Number

Step 1: Filter townships

**COOK COUNTY**
GOVERNMENT | OPEN
DATA

Search

Home Browse All Data Cook County Website About Us User Resources ▾ Contact Us ▾

[← Back to overview](#) [↶](#) [↷](#) Search

 keypin keypin	 pins pins	# year year	 township township	
16-08-117-007-0000	16-08-110-017-0000, 16-08-110-018-0...	2023	Oak Park	T27-S
16-07-300-013-0000	16-07-300-013-0000	2023	Oak Park	T27-M
16-07-128-002-0000	16-07-128-002-0000, 16-07-128-031-0...	2023	Oak Park	T27-M
16-07-120-065-0000	16-07-120-065-0000	2023	Oak Park	T27-M
16-18-201-032-0000	16-18-200-005-0000, 16-18-201-032-0...	2023	Oak Park	T27-M

 Filters | [✕ Clear all](#)



Search...

River Forest ✕ Oak Park ✕

Step 2: Add a filter, for bldg. sf

The image shows a user interface for configuring filters. A sidebar on the left contains a 'Filters' section with a 'Clear all' button and a list of existing filters. The main area displays a filter configuration for 'township' with the condition 'is one of'. A new filter is being added, indicated by a '+' button and the word 'AND'. The new filter is for '# bldgsf' with the condition 'is greater than or equal to' and the value '100000'. The 'Apply' button is highlighted with a red circle.

Filters | [Clear all](#)

Tt township [is one of](#)

+ AND

AND

bldgsf [is greater than or equal to](#) 100000

+ AND

[Apply](#)

Step 3: Review and export

[← Back to overview](#)



[Export](#)

 keypin <i>keypin</i>		 pins <i>pins</i>		# year <i>year</i>		 township <i>township</i>		 modelgroup <i>sheet</i>		
16-08-117-007-0000		16-08-110-017-0000, 16-08-110-018-0...		2023		Oak Park		T27-SpecialMultiClass		5-90,
16-07-300-013-0000		16-07-300-013-0000		2023		Oak Park		T27-Multifamily		3-97
16-07-128-002-0000		16-07-128-002-0000, 16-07-128-031-0...		2023		Oak Park		T27-Multifamily		3-97,
16-07-120-065-0000		16-07-120-065-0000		2023		Oak Park		T27-Multifamily		3-91
16-18-201-032-0000		16-18-200-005-0000, 16-18-201-032-0...		2023		Oak Park		T27-Multifamily		3-97
16-07-118-060-0000		16-07-118-060-0000, 16-07-118-066-0...		2023		Oak Park		T27-Multifamily		3-97,
16-08-126-003-0000		16-08-126-003-0000, 16-08-126-004-0...		2023		Oak Park		T27-Multifamily		3-91,
15-01-418-022-0000		15-01-418-022-0000		2023		River Forest		T33-Multifamily		3-97
16-18-110-015-0000		16-18-110-006-0000, 16-18-110-015-0...		2023		Oak Park		T27-SpecialMultiClass		5-90,
15-12-220-038-0000		15-12-220-038-0000		2023		River Forest		T33-SpecialMultiClass		4-97
16-08-117-015-0000		16-08-117-015-0000, 16-08-117-016-0...		2023		Oak Park		T27-SpecialMultiClass		5-91

Showing rows 1-13 of 13

 Filters |  Clear all



Tips and suggestions

Navigating the labyrinthine data model of property tax assessment.

Cook County Open Data Portal

General suggestions

Check the docs!

Every Open Data asset has metadata that describes the data and provides helpful context.

Assessor - Commercial Valuation Data

- The Assessor's Office reassesses roughly one third of the county (a triad) each year. For commercial valuations, this means each year of data only contain the triad that was reassessed that year.
- One KeyPIN is one Commercial Entity.
- Each KeyPIN (entity) can be comprised of one single PIN (parcel), or multiple PINs as designated in the pins column.

Cook County Open Data Portal

General suggestions



Excel issues

Excel is limited to ~1M rows.
Cook County has roughly 1.9M parcels.

If you download **all PINs** in a tax year, excel will **not** accurately show or analyze your data.



Excel solutions

Options include:

- Filter your data to reduce rowcount. Class code is a common way to filter for only residential parcels.
- Use google sheets, if allowed. It has a cell cap of 10M. With 1 column of data, you can have 10M rows; and with 2 columns, 5M; so on.

Cook County Open Data Portal

General suggestions



Joining datasets

Frequently, users have questions that require combining datasets.

Example: “For **Permit PINs**, I want to know their **Assessed Values**.”



General steps

1. Determine the PINs you’re interested in. Download that list.
2. Download additional data that you want to attach to those PINs.
3. Open your first list, and join it to your second list by PIN.

Or: ask us for help!

Cook County Open Data Portal

General suggestions



What's a Tax Year?

“Tax Year” defines the fiscal tax bill calculation and can be different from the calendar year.

We are in **calendar** year 2026. We are conducting 2026 revaluations, estimating property values in 2026.

This will affect tax bills for Tax Year 2026, which will be issued in **calendar** year 2027.



Year explanation

Most of our datasets rely on accurate application of “Tax Year”.

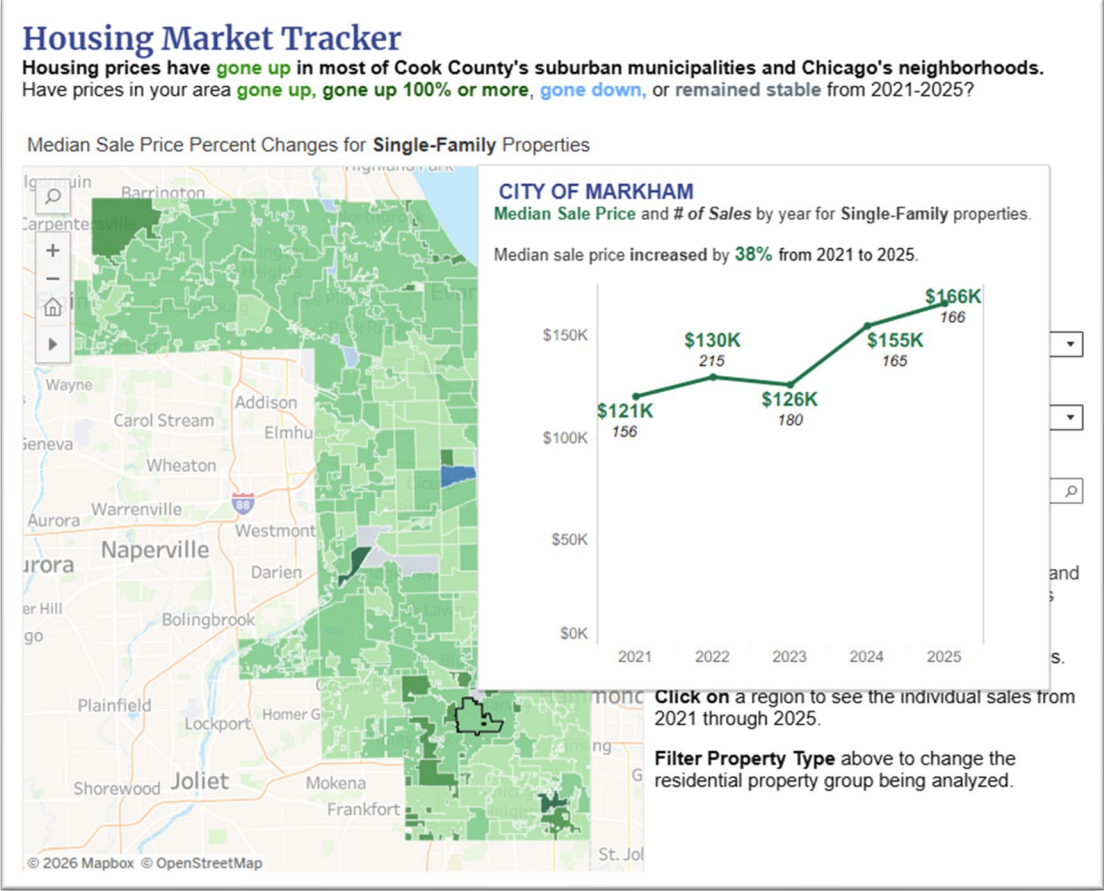
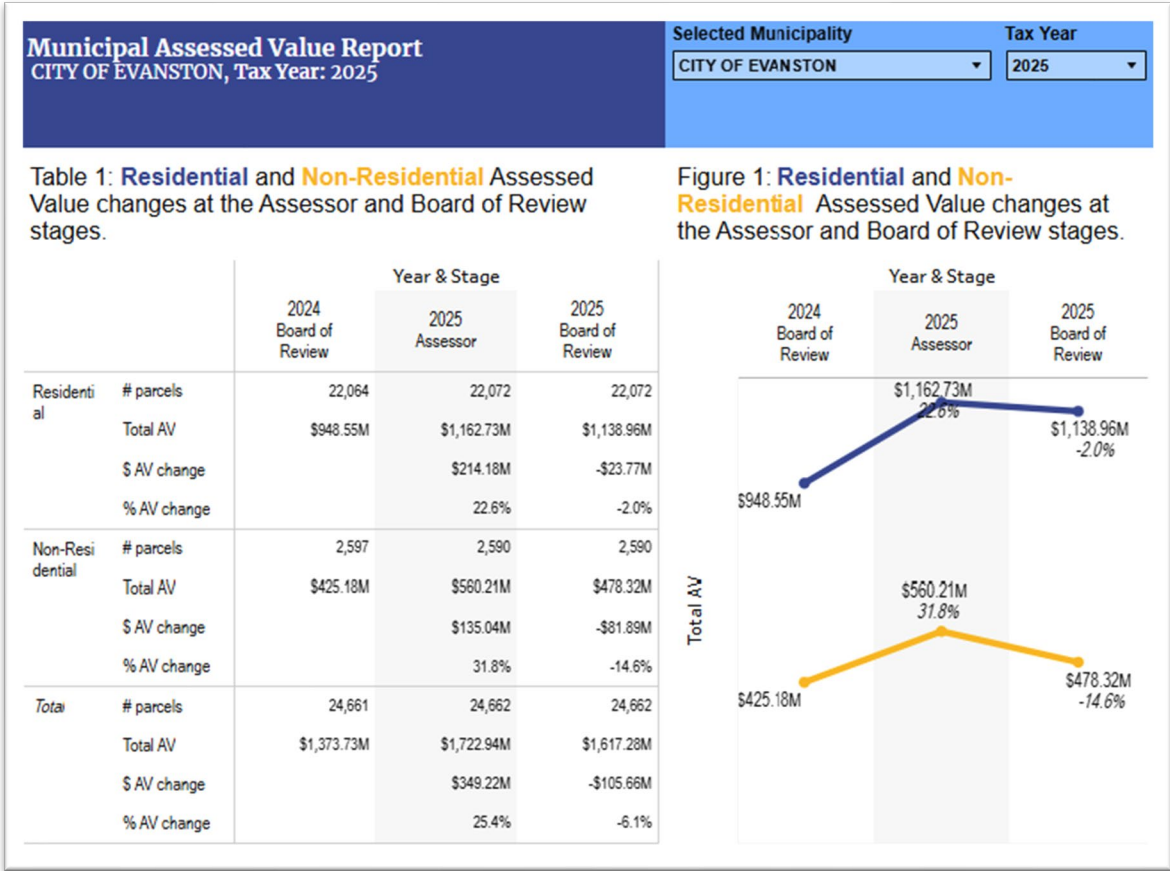
When you join two datasets together, make sure you're using the same tax year.



Bonus: Dashboards

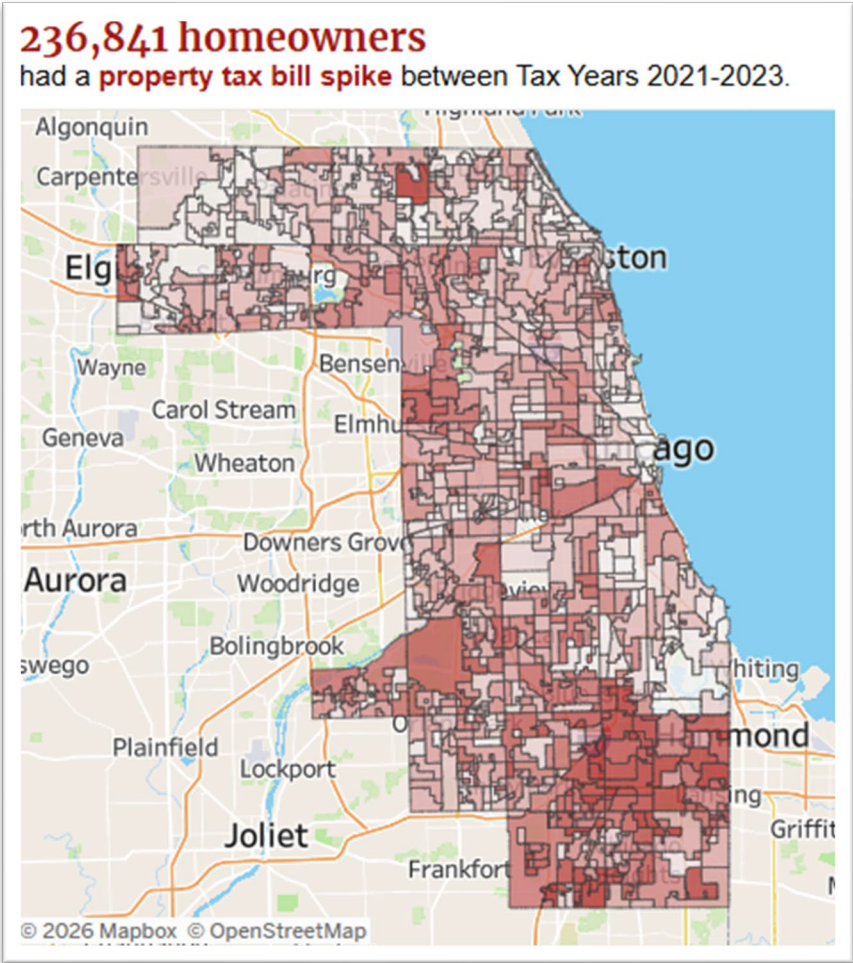
Dashboards, with downloadable data

cookcountyassessoril.gov/dashboard



Dashboards, with downloadable data

cookcountyassessoril.gov/dashboard



Impact: Affordable Housing Special Assessment Program

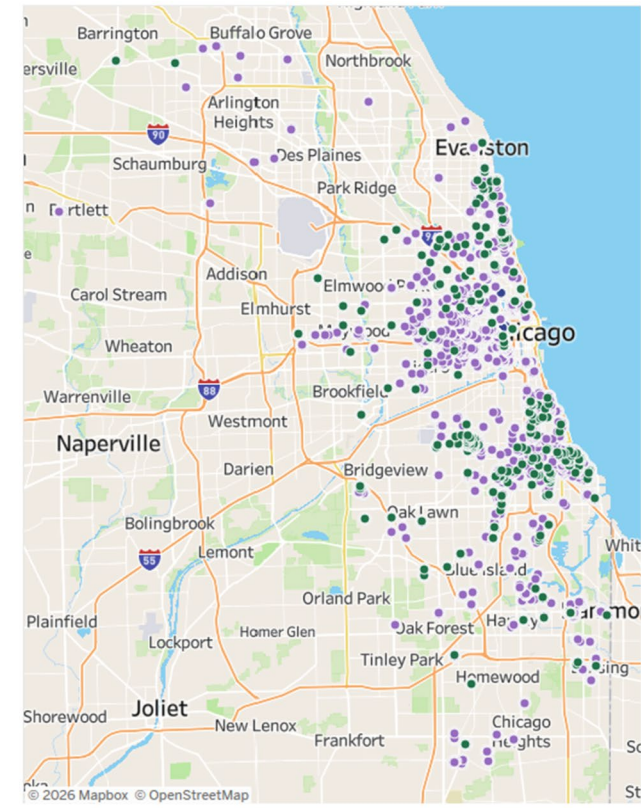
In Tax Year 2022, the Cook County Assessor started to administer the Affordable Housing Special Assessment Program (AHSAP). How many projects have joined the program through 2025?

Projects by Tier
Select tier to filter dashboard data

Tier 15	Tier 35	LAC	Grand Total
260 projects	929 projects	6 projects	1,195 projects
395 PINs	2,028 PINs	31 PINs	2,454 PINs

Projects by Year Accepted
Select year to filter dashboard data

2022	2023	2024	2025	Grand Total
487 projects	209 projects	346 projects	153 projects	1,195 projects
1,049 PINs	355 PINs	673 PINs	377 PINs	2,454 PINs





Questions? Requests?

Assessor.Data @ cookcountyil.gov

github.com/ccao-data

Appendix

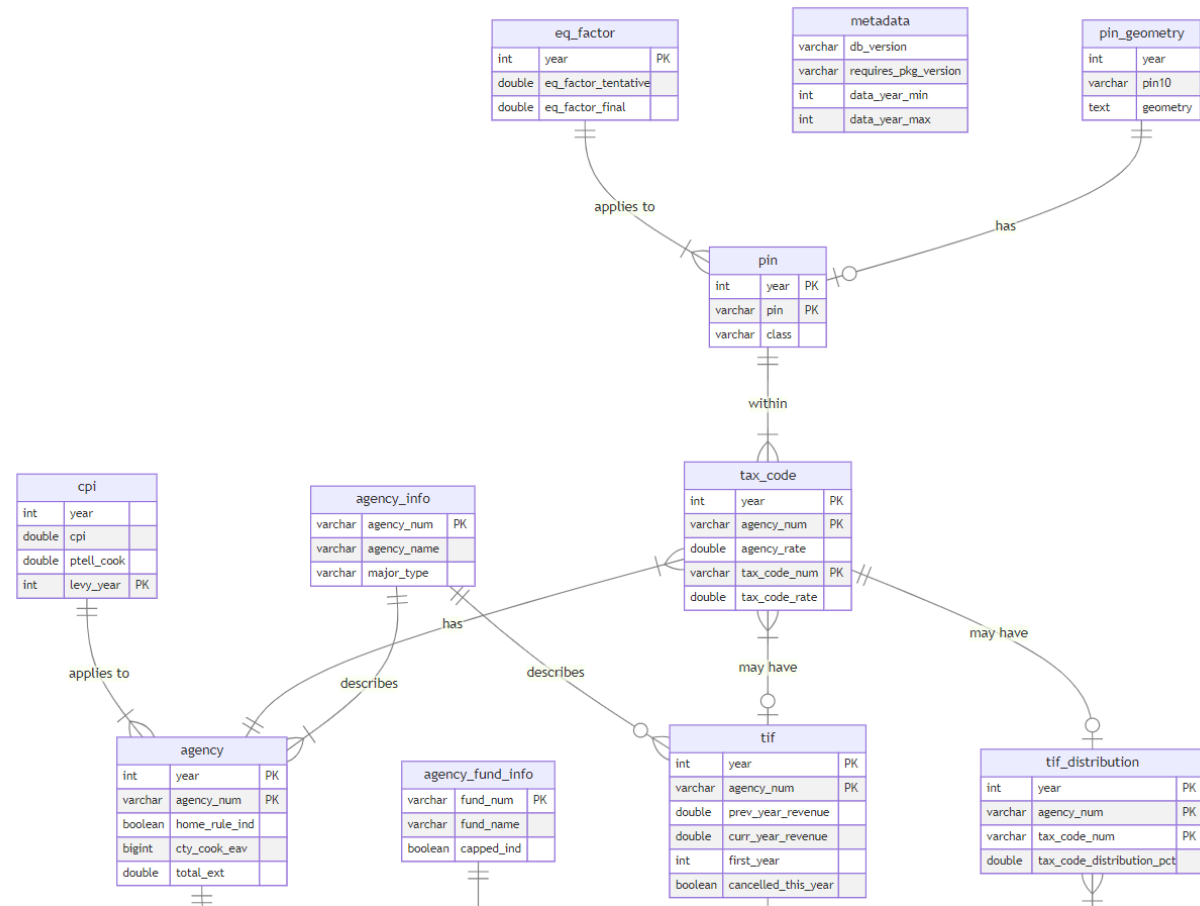


Bonus: PTAXSIM

PTAXSIM

<https://github.com/ccao-data>

- **database** of line items for property tax levies and bills, 2006-2024
- **code** for property tax analysis and simulation



PTAXSIM

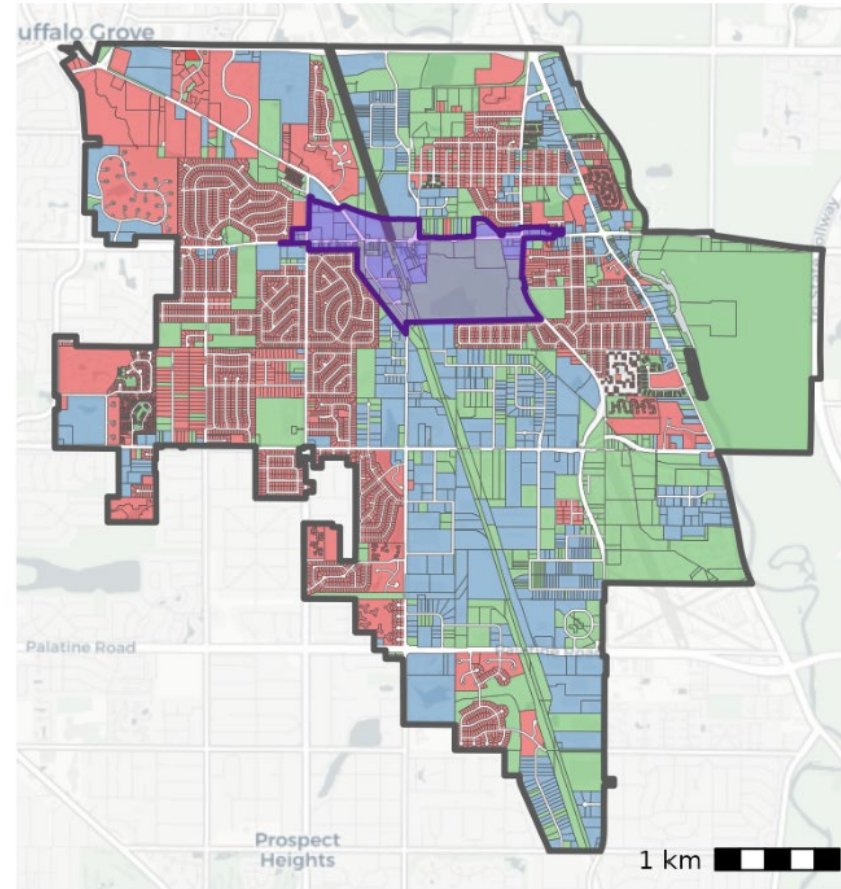
<https://github.com/ccao-data>

- **articles** that demonstrate how to do frequently requested analyses

```
# PIN geometries only exist for 10-digit PINs, so we'll need to truncate our
# existing Wheeling PINs
wh_pin10s <- unique(substr(wh_pins$pin, 1, 10))

# The geometry is stored in the database as simple text, so we need to convert
# it to the correct type using sf
wh_pins_geo <- lookup_pin10_geometry(year = 2020, pin10 = wh_pin10s)
wh_pins_geo <- wh_pins_geo %>%
  st_as_sf(wkt = "geometry", crs = 4326)

# We'll also attach the first class code from each PIN14 to each PIN10 geometry
# and group them into broad buckets. This is just to help our visualization
wh_pins_geo <- wh_pins_geo %>%
  left_join(
    wh_pins %>%
      mutate(pin10 = substr(pin, 1, 10)) %>%
      group_by(pin10) %>%
      summarize(class = first(class)),
    by = "pin10"
  ) %>%
  mutate(
    major_class = substr(class, 1, 1),
    major_class_fct = recode_factor(
      major_class,
      "2" = "2 - Residential",
      "3" = "3 & 5 - Commercial",
      "5" = "3 & 5 - Commercial",
      "0" = "Other", "1" = "Other", "6" = "Other", "7" = "Other"
    )
  )
```



2 - Residential 3 & 5 - Commercial Other